Module 07: Communication in High School

Learning Objectives

1. Define **Communication** within the context of High School.
2. Analyze how Communication interacts with other components of the Active Inference framework.
3. Apply specific constraints of High School to the formal definition of Communication.

Introduction

This module explores **Communication**. In the **High School** curriculum, we approach this topic with a focus on specific applications and theoretical depth appropriate for the audience. Communication is a critical component of the 8-part Active Inference spine, bridging the gap between Learning and Planning.

Key Concepts

1. Communication as a Markov Blanket Boundary

How does Communication define the boundary between the agent and the environment?

2. Generative Models of Communication

What parameters involved in Communication must be optimized to minimize variational free energy?

3. Active Inference Dynamics

How does the process of Communication drive the perception-action loop?

Applications

In High School, we see Communication manifest in: * **Specific Example 1**: Shannon's information theory provides the mathematical foundation for communication in Active Inference: the information content of a message is $I = -\log(p)$, meaning rare (surprising) messages carry more information than expected ones -- this is why "School is cancelled due to snow" (low probability, high surprise) is a more informative communication than "School starts at 8 AM" (high probability, low surprise), and quantifies exactly how much a message updates the receiver's generative model. * **Specific Example 2**: Error-correcting codes like Hamming codes demonstrate mathematical communication through redundancy: by adding parity check bits to a binary message, the receiver can detect and correct single-bit errors -- mathematically, the code embeds the message in a higher-dimensional space where prediction errors (corrupted bits) can be identified and fixed, ensuring the received signal matches the sender's intended generative model.

Conclusion

Understanding Communication allows us to better model complex adaptive systems. In the next module, we will expand on this foundation.